An Exact Major-Arc Gate for Additive Phases: Periodic Projection and a Minor-Arc Route Obstruction

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Abstract

TPC-157 transfers a periodic two-Möbius estimate to a literal multiplier at the cost of its normalized L^1 distance from a small-period subspace. We evaluate that gate for an additive phase. A phase within inverse-interval accuracy of an allowed rational has a phase-aligned periodic approximant and inherits logarithmic cancellation. On a complete residue rectangle, the orthogonal projection error is an exact Dirichlet-kernel expression; away from all allowed rationals it is bounded below, stopping the periodic approximation route. The obstruction concerns the approximation method, not the size of the original Möbius correlation.

Keywords. additive phase; major arcs; Dirichlet kernel; periodic approximation; Möbius correlation.

Claim boundary. The positive theorem covers an explicit major-arc cell of the actual-core sum. The negative theorem says that a specified small-period approximation is expensive; it does not say that the minor-arc correlation is large or that every route fails. The production phase is not yet supplied by an occurrence registry. There is no all-prefix, positive fixed-X-power, 1/400, prime-pair or twin-prime conclusion.

1 Setup

Write $\mathrm{e}(x) = \exp(2\pi i x)$. Retain the determinant-two notation

$$q = as, \qquad t(z) = ad + qz, \qquad c_z = \mu(d + sz)\mu(u + az), \qquad I_N = \{z : N < t(z) \le 2N\}.$$

TPC-149 gives constants $\eta_0, \kappa_0 > 0$ and, for every sufficiently large X, one set $\mathcal{E}_X^\star \subset [\sqrt{X}, X]$ outside which the simultaneous estimate

$$\frac{q}{N}\left|\sum_{z\in I_N} c_z\rho(z)\right| \ll \|\rho\|_\infty(\log X)^{-\kappa_0} \tag{1}$$

holds for every period-R function with $qR \le (\log X)^{\eta_0}$ [1]. TPC-157 adds the normalized L^1 approximation error for a nonperiodic multiplier [2].

2 The major-arc gate

Let $I = \{z_0, z_0 + 1, \ldots, z_0 + L - 1\}$ contain I_N; enlarging the error sum outside I_N is harmless.

Lemma 2.1 (Phase-aligned approximant)**.** *For $\alpha \in \mathbb{R}$, $k \in \mathbb{Z}$, and $R \geq 1$, put*

$$\rho_{z_0}(z) = \mathrm{e}(-\alpha z_0)\,\mathrm{e}\Big(-\frac{k}{R}(z - z_0)\Big).$$

Then ρ_{z_0} is period R, has modulus one, and

$$\sup_{z \in I} |\mathrm{e}(-\alpha z) - \rho_{z_0}(z)| \leq 2\pi L \left|\alpha - \frac{k}{R}\right|. \tag{2}$$

Proof. Periodicity follows from $\mathrm{e}(-k) = 1$. After removing the common factor $\mathrm{e}(-\alpha z_0)$, use $|\mathrm{e}(x) - \mathrm{e}(y)| \leq 2\pi|x - y|$ and $|z - z_0| \leq L$. ☐

Theorem 2.2 (Actual-core major arc)**.** *Fix $A > 0$. Suppose*

$$qR \leq (\log X)^{\eta_0}, \qquad L\left|\alpha - \frac{k}{R}\right| \leq (\log X)^{-A},$$

and

$$N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^{\star}.$$

Then

$$\boxed{\frac{q}{N}\left|\sum_{z \in I_N} c_z \mathrm{e}(-\alpha z)\right| \ll (\log X)^{-\kappa_0} + (\log X)^{-A}.} \tag{3}$$

Proof. Use ρ_{z_0} in the TPC-157 interface. The periodic term gives the first summand. Since $|I_N| \ll N/q + 1$, (2) gives the second, after absorbing the harmless endpoint term in the asymptotic range. ☐

The alignment at z_0 matters. Comparing directly with $\mathrm{e}(-kz/R)$ would incorrectly charge the absolute coordinate rather than the interval length.

3 Exact projection on a residue rectangle

Let

$$I_{K,R} = \{z_0, \ldots, z_0 + KR - 1\}, \qquad f_\alpha(z) = \mathrm{e}(\alpha z),$$

and let $\mathcal{P}_R$ be the space of period-R functions restricted to this interval.

Theorem 3.1 (Dirichlet projection identity)**.** *For every $K, R \geq 1$,*

$$\frac{1}{KR} \inf_{\rho \in \mathcal{P}_R} \sum_{z \in I_{K,R}} |f_\alpha(z) - \rho(z)|^2 = 1 - \left|\frac{1}{K}\sum_{j=0}^{K-1} \mathrm{e}(\alpha R j)\right|^2. \tag{4}$$

Equivalently, with the continuous value 1 for the ratio when $R\alpha \in \mathbb{Z}$, the squared mean on the right is

$$\left|\frac{\sin(\pi K R \alpha)}{K \sin(\pi R \alpha)}\right|^2.$$

Moreover,

$$\frac{1}{KR} \inf_{\rho \in \mathcal{P}_R} \sum_{z \in I_{K,R}} |f_\alpha(z) - \rho(z)| \geq \frac{1}{2}\left[1 - \left|\frac{1}{K}\sum_{j=0}^{K-1} \mathrm{e}(\alpha R j)\right|^2\right]. \tag{5}$$

Proof. On each residue fiber $z = z_0 + r + jR$, the orthogonal projection is the fiber mean

$$\mathrm{e}(\alpha(z_0 + r))\frac{1}{K}\sum_{j=0}^{K-1} \mathrm{e}(\alpha Rj).$$

Pythagoras gives (4), and summing the geometric progression gives the sine quotient. An L^1 minimizer may be projected into the closed unit disk without increasing distance from unit-modulus data. There $|f_\alpha - \rho| \le 2$, hence $|f_\alpha - \rho|^2 \le 2|f_\alpha - \rho|$. Infimizing proves (5). □

Corollary 3.2 (Uniform minor-arc route stop)**.** *Let I_n be cells of L_n consecutive integers, let $\alpha_n \in \mathbb{R}$, and let $\mathcal{R}_n$ be a nonempty set of allowed periods. Put*

$$K_{n,R} = \left\lfloor \frac{L_n}{R} \right\rfloor \qquad (R \in \mathcal{R}_n).$$

If

$$\inf_{R\in\mathcal{R}_n} K_{n,R} \longrightarrow \infty, \qquad \inf_{R\in\mathcal{R}_n} K_{n,R}\|R\alpha_n\|_{\mathbb{R}/\mathbb{Z}} \longrightarrow \infty, \tag{6}$$

then, uniformly over all allowed periods,

$$\inf_{R\in\mathcal{R}_n} \frac{1}{L_n} \inf_{\rho \text{ period } R} \sum_{z\in I_n} |f_{\alpha_n}(z) - \rho(z)| \ge \frac{1}{2} - o(1).$$

Consequently, the TPC-157 small-period approximation route cannot yield a decaying bound on those cells.

Proof. For each $R \in \mathcal{R}_n$, retain a complete $K_{n,R}R$-subinterval of I_n. Restriction and (5) give the lower bound

$$\frac{K_{n,R}R}{L_n}\frac{1}{2}\left[1 - \min\left\{1, \frac{1}{2K_{n,R}\|R\alpha_n\|_{\mathbb{R}/\mathbb{Z}}}\right\}^2\right].$$

The first infimum in (6) makes $K_{n,R}R/L_n = 1 + o(1)$ uniformly, and the second makes the squared term $o(1)$ uniformly. □

The uniform infima in (6) are essential. A pointwise assertion for every fixed R does not stop a sequence of periods $R = R_n$ drifting with n.

4 Route semantics

The positive result is labeled L1_ACTUAL_CORE_MAJOR_ARC. The negative result is a scoped route stop:

$$\text{small-period approximation fails} \not\Rightarrow \text{the Möbius correlation is large.}$$

A direct additive-twist theorem, a larger admissible period range, or a different physical crosswalk remains logically open. The accompanying audit checks the projection identity numerically against the direct orthogonal projection and checks a nonzero-start phase-aligned fixture. These finite checks support implementation; the proofs above establish the identities.

5 Conclusion

The phase question now has a sharp gate. Near an allowed rational, the existing actual-core theorem genuinely advances to an additive phase. Away from all allowed rationals, the exact Dirichlet projection shows that periodic approximation cannot be the missing argument. This narrows the next arithmetic input without pretending to resolve generic phases.

References

[1] Liang Wang. A Small-Polylogarithmic Determinant-Two Möbius Corridor, 2026. TPC-149 manuscript.

[2] Liang Wang. Periodic Approximation of Literal Multipliers on Determinant-Two Möbius Fibers, 2026. TPC-157 manuscript.